

Module 04: Cognition — Precision Weighting, Neuromodulation, and Attention

Learning Objectives

1. Explain how **precision weighting** — the gain control on prediction errors — implements attention in the brain.
2. Describe the neuromodulatory systems (dopamine, serotonin, noradrenaline, acetylcholine) that regulate precision and their roles in cognition.
3. Analyze ADHD, autism, and anxiety as specific patterns of aberrant precision weighting.
4. Distinguish between **content precision** (confidence in specific predictions) and **context precision** (confidence in the model structure).

Introduction

Cognition in the Active Inference framework is not a separate faculty but the result of **precision weighting** — the brain's mechanism for determining which prediction errors matter and which can be safely ignored. This is the neural implementation of attention: high precision amplifies a signal; low precision attenuates it. The neuromodulators (dopamine, acetylcholine, serotonin, noradrenaline) are the brain's "precision engineers," adjusting the gain on prediction errors across the entire hierarchy.

Key Concepts

1. Precision as Neural Gain

In mathematical terms, precision is the inverse variance ($\Pi = 1/\sigma^2$) of a probability distribution. Neurally, precision is implemented as the **gain** on prediction error units — how strongly a given prediction error drives updates to the generative model. High-precision prediction errors demand updating; low-precision prediction errors are ignored.

Feldman and Friston (2010) proposed that precision is encoded by the synaptic gain of superficial pyramidal cells — the same neurons that carry prediction errors up the cortical hierarchy. Neuromodulators control this gain: **acetylcholine** increases precision on bottom-up sensory prediction errors (making the brain more sensitive to sensory data); **dopamine** modulates precision on predictions about reward and policy outcomes.

2. Neuromodulatory Systems and Precision

Neuromodulator	Primary Precision Role	Clinical Relevance
Acetylcholine	Increases precision on sensory (bottom-up) prediction errors	Basal forebrain degeneration (Alzheimer's) → reduced sensory precision
Dopamine	Modulates precision on reward predictions and policy selection	Excess → psychosis (overweighting predictions); deficit → Parkinson's akinesia
Noradrenaline	Controls overall arousal and the volatility of the environment model	Dysregulation → PTSD (hyperarousal), attention disorders
Serotonin	Regulates affective precision and interoceptive prediction errors	Imbalance → depression, anxiety

3. Attention as Active Inference

Attention is not a separate mechanism bolted onto perception — it **is** precision optimization. Parr and Friston (2017) showed that attentional selection can be fully explained as the brain optimizing precision on specific sensory channels:

- **Endogenous (top-down) attention:** Increasing precision on sensory channels predicted to be informative. When searching for a friend in a crowd, the brain upweights face-related prediction errors and downweights background texture errors.

- **Exogenous (bottom-up) attention:** Large, unexpected prediction errors automatically receive high precision. A loud bang captures attention because the prediction error is high-amplitude and demands model updating.
- **Sustained attention:** Maintaining high precision on task-relevant channels over time, which requires continuous engagement of the prefrontal-parietal network to counteract the brain's default tendency to explore (switch to novel sources of prediction error).

4. Clinical Applications

- **ADHD:** May reflect low tonic dopamine → difficulty sustaining precision on task-relevant prediction errors → constant shifting of attention to novel (high prediction error) stimuli.
- **Autism Spectrum:** Lawson et al. (2014) proposed that autism involves high precision on sensory prediction errors → difficulty ignoring sensory details → reduced influence of top-down contextual priors, leading to both perceptual strengths (pattern detection) and challenges (sensory overwhelm, difficulty with ambiguity).
- **Anxiety:** Excessive precision on threat-related prediction errors → a generative model chronically biased toward detecting danger, even in safe environments.
- **Depression:** Reduced precision on positive prediction errors and interoceptive allostatic signals → the brain treats positive outcomes as uninformative, maintaining a pessimistic generative model.

5. Metacognition and Confidence Monitoring

Precision is not only applied to first-order predictions but also to **second-order beliefs about confidence**. Metacognition — the ability to monitor one's own cognitive states — requires a higher-order model that estimates the precision of lower-level inferences. Fleming et al. (2010) showed that metacognitive accuracy correlates with the volume of anterior prefrontal cortex (BA10), suggesting a dedicated neural substrate for hierarchical precision estimation.

Conclusion

Precision weighting is the bridge between the abstract mathematics of Active Inference and the neurochemistry of the brain. By understanding how neuromodulators control the gain on

prediction errors, we connect theory to pharmacology, clinical psychiatry, and the everyday experience of attention. Module 05 examines how precision-weighted inference drives action and motor control.

Further Reading

- Feldman, H. & Friston, K. (2010). Attention, uncertainty, and free-energy. *Frontiers in Human Neuroscience*, 4, 215.
- Parr, T. & Friston, K. J. (2017). Working memory, attention, and salience in active inference. *Scientific Reports*, 7, 14678.
- Lawson, R. P. et al. (2014). An aberrant precision account of autism. *Frontiers in Human Neuroscience*, 8, 302.